



DEPARTMENT OF THE AIR FORCE
HEADQUARTERS UNITED STATES SPACE FORCE

14 February 2024

MEMORANDUM FOR 5CXXX CAREERFIELD

FROM: HQSF/COO/5C CAREER FIELD MANAGER (CFM)

SUBJECT: Career Field Education and Training Plan (CFETP) Guidance

1. As we continue to work towards 5CXXX specific CFETPs we realize that further guidance is needed in regards to upgrade training for our Cyber Guardians. Previously, a guidance memorandum was published on 30 Oct 2023 directing 5C's to follow specific tasks within the USAF 1D7 CFETP. Since 30 Oct 2023, the 5CXXX, 5C0X1D, and 5C0X1R CFETP part II's have been published. This memorandum, along with the attached Part II's, is intended to provide the field with accurate tasks and guidance which should be considered for upgrade training.

2. Until further policy and guidance for 5C0X1's is finalized, we will continue to use the information in this memorandum, the attached, and sections of the 1D7XX/X CFETP listed below for all upgrade training and education and training requirements for skill levels.

a) CFETP 1D7XX/X, dated 31 July 2023, can be found at the below link:

i) <https://www.e-publishing.af.mil/product-index/#/?view=search&keyword=1D7&isObsolete=false&modID=449&tabID=131>

3. Each shred will complete the below tasks, to include rank requirements, in addition to any locally determined training requirements: e.g. Work center Job Qualification Standards, Initial Qualification Training, Missions Qualification Training, etc.

- a) 5CXXX's will complete local requirements and tasks in Attachment 1, 5CXXX CFETP Part II.
- b) 5C0X1D's will complete local requirements and tasks in Attachment 2, 5C0X1D CFETP Part II.
- c) 5C0X1R's will complete local requirements and tasks in Attachment 3, 5C0X1R CFETP Part II.
- d) 5C0X1N/S's will complete local requirements and tasks 2-3 & 5-23 in Attachment 4, 1D7X1Q.

4. In addition to specific tasks associated with 5C shreds, the below outlines further education and training requirements for the award of 3/5/7-Skill Level.

a) 3-Skill Level

i) Awarded upon completion of 3-Skill Level Initial Skills Training (RF Transmissions Operations course, Network Systems Operations course, Systems Operations course, or Cyber Warfare Operator course).

b) 5-Skill Level

i) Minimum Rank: Specialist 3

ii) No minimum required time-in-training, Maximum time-in-training is 24 months.

c) 7-Skill Level

- i) Minimum Rank: Sergeant
- ii) No minimum required time-in-training, Maximum time-in-training is 24 months.

5. Training completion will be tracked through locally determined methods until myTraining becomes fully functional. Coordination with Unit Training Managers is essential to ensure Guardian's records are accurate.

6. This memorandum supersedes the previous memorandum, same subject, dated 30 Oct 2023. Direct questions concerning this memorandum to the Space Force Enlisted Cyber CFM at (703) 693-5807 or via email at stephen.shockey@spaceforce.mil.

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STEPHEN M. SHOCKEY, SMSgt, USSF
Enlisted Cyber Career Field Manager

Qualitative Requirements

PROFICIENCY CODE KEY		
	SCALE VALUE	DEFINITION: The individual
Task Performance Levels	1	Can do simple parts of the task. Needs to be told or shown how to do most of the task. (EXTREMELY LIMITED)
	2	Can do most parts of the task. Needs help only on hardest parts. (PARTIALLY PROFICIENT)
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*Task Knowledge Levels	a	Can name parts, tools, and simple facts about the task. (NOMENCLATURE)
	b	Can determine step by step procedures for doing the task. (PROCEDURES)
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**Subject Knowledge	A	Can identify basic facts and terms about the subject. (FACTS)
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Explanations * A task knowledge scale value may be used alone or with a task performance scale value to define a level of knowledge for a specific task. (Example: b and 1b) ** A subject knowledge scale value is used alone to define a level of knowledge for a subject not directly related to any specific task, or for a subject common to several tasks. This mark is used alone instead of a scale value to show that no proficiency training is provided in the course or Learning Program. (-) This mark is used alone in the Proficiency Codes Course columns to show no proficiency training is provided in the applicable course. Training is satisfied through OJT, CBTs, Learning Programs, or a combination. (X) This mark is used alone in the Proficiency Codes Course columns to show training is required but not given due to limitations in resources or is a future requirement. Training is satisfied through OJT, CBTs, Learning Programs, or a combination. NOTE: All tasks and knowledge items shown with a proficiency code are trained during wartime. (X) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task as a core requirement to the level identified in the Task Knowledge Levels Column. The training to satisfy this requirement is either provided through OJT, CBTs, Learning Programs, or a combination. (*) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task as a wartime/deployment requirement. (^) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task require(s) third person certification.		

CFETP versus AFJQS task coding. AFJQSs/AFQTPs annotated in the CFETP with a skill level denotes the AFJQS is mandatory.

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
1. CYBER DEFENSE OPERATIONS TR: 1D7XX Learning Program (Percipio); AFMAN 36-2100; AFPD 172; 5CXXX CFETP; AFECD; JPs 3-12, 3-13, 3-13.1, and 3-13.2										
1.1. Introduction to Cyber										
1.1.1. Cyber Operations Roles and Responsibilities	5/7/E						A	A	A	C
1.1.2. Concepts of FMA / OCO / DCO / DoDIN Ops	5/7/E						A	A	A	C
1.1.3. Qualifications	5/7						-	A	A	-
1.1.4. Progression within SFSC	5/7						-	A	B	-
1.1.5. Read CFETP 5CXXX, Part I	5/7						-	A	A	-
2. PUBLICATIONS AND DIRECTIVES TR: 1D7XX Learning Program (Percipio), AFI 33-360, TO 00-5-1, https://dtic.mil										
2.1. Publications	5/7						A	A	B	-
2.2. Technical Orders (TO)	5/7						A	A	A	-
2.3. Locate Applicable Publications	5/7						-	2b	2b	-
2.4. Use Publications / Technical Orders When Performing Work	5/7						2b	b	b	-
3. CYBER AUTHORITY AND USER RESPONSIBILITIES TR: 1D7XX Learning Program (Percipio); AFDP 3-13 https://www.doctrine.af.mil/Doctrine-Annexes/Annex-3-13-Information-Ops/ ; JP 3-13; USC Titles 10, 18, 50; http://iase.disa.mil/eta/Pages/index.aspx										
3.1. Anti-Piracy	5						A	A	-	-
3.2. Ethics	5/7/E						A	A	B	D
3.3. US Codes	5/7						-	A	B	C
3.4. US Telecommunications Laws	7/E						-	-	A	B
3.5. International Laws Affecting Electronic Communications	7/E						-	-	A	B
3.6. LOAC Considerations When Planning / Conducting Cyber Operations	7/E						-	-	A	B
3.7. Rules of Engagement (ROE)	5/7/E						A	B	B	B
3.8. Cyber Tasking Order (CTO)	5/7/E						-	A	B	C
4. CYBER ENCLAVE TR: 1D7XX Learning Program (Percipio), AFD 17-2, AFI 17-201, JP 3-12, http://www.afcyber.af.mil										
4.1. Structure	5/7/E						-	A	B	B
4.2. Missions	5						-	A	-	-
4.3. Offensive Cyberwarfare Operations	5/7/E						-	A	B	C
4.4. Defensive Cyberwarfare Operations	5/7/E						-	A	B	C
4.5. Exploitation	5/7						-	A	A	-
4.6. Effects on Adversary Decision Makers	5/7/E						-	B	B	C

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.7. Units										
4.7.1. Joint, DOD, and Combined Units	5/7						-	A	B	-
4.7.2. Space Force Units	5/7						-	A	B	-
4.7.3. Air Force Network Operations (AFNetOps)	5/7						-	B	B	-
4.7.4. Cyber Organizations (JCSE, WHCA, DISA, etc.)	5/7						A	A	B	-
5. ENTERPRISE SYSTEMS / PROGRAMS TR: 1D7XX Learning Program (Percipio); AFI 13-Series; Joint Pub 6-0; CJCSIs 3231.01C, 6211.02D										
5.1. Define Non-Secure Networks	5						A	A	-	-
5.2. Define Secure Networks	5						A	A	-	-
5.3. Nuclear Command, Control, and Communications Systems	5						-	A	-	-
5.4. Space Systems	5						-	A	-	-
5.5. Airborne Networks	5/7						-	A	A	-
5.6. Battlefield Networks	5/7						-	A	A	-
6. SAFETY / RISK MANAGEMENT (RM) TR: 1D7XX Learning Program (Percipio); AFIs 32-1065, 90-802, 91-202; AFMAN 91-203; AFPAM 90-803; AFD 91-2; MIL-STD 188-124B										
6.1. Safety	5						-	B	-	-
6.2. RM	5/7						-	A	B	-
6.3. Work Center Safety Program Management	5/7						-	A	A	-
6.4. Characteristics of Personal and Equipment Protection	5						-	A	-	-
7. CYBER SECURITY TR: 1D7XX Learning Program (Percipio); AFIs 10-701, 17-130, 16-1404; AFD 10-7; AFMAN 17-1301										
7.1. Risk, Threats, and Vulnerabilities	5/7/E						A	A	B	C
7.2. Network Security	5/7						B	A	B	-
7.3. Firewalls										
7.3.1. Concepts	5						A	A	-	-
7.4. Security Zones	5						-	A	-	-
7.5. Incident Response	5						A	A	-	-
7.6. Cyber Hygiene										
7.6.1. Acceptable Use / Behavior for Information Technology	5						A	A	-	-
7.7. Security Technical Implementation Guides (STIGs)										
7.7.1. STIG Fundamentals	5						A	B	-	-
7.7.2. STIG Principles	5						A	B	-	-
8. NETWORK FUNDAMENTALS TR: 1D7XX Learning Program (Percipio)										
8.1. IP Addressing Concepts	5						A	B	-	-
8.2. OSI / TCP / IP Models	5/7						B	B	B	

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
8.3. Network Devices	5/7						A	A	A	-
8.4. Communications Mediums	5/7						A	A	A	-
8.5. LAN Architecture	5/7						A	A	A	-
8.6. Ports, Protocols, and Services (PPS)	5/7						B	B	B	-
8.7. Apply IP Address	5						2b	B	-	-
8.8. Network Types	5/7						A	B	B	-
8.9. Networks Layers	5						A	A	-	-
8.10. Subnetting	5						A	B	-	-
8.11. Network Topologies										
8.11.1. Topologies Fundamentals	5/7						A	B	B	-
8.11.2. Develop Logical Diagram	5						2b	2b	-	-
8.11.3. Develop Physical Diagram	5						2b	2b	-	-
8.11.4. Develop Data Flow Diagram	5						2b	2b	-	-
8.12. Network Monitoring	5						A	B	-	-
8.13. Wireless Networking (WLAN)	5/7						B	B	B	-
8.14. Backup and Restore Device Documentation	5						A	A	-	-
8.15. Troubleshooting Methodology	5						B	B	-	-
8.16. Cable Management										
8.16.1. Cable Types and Handling	5						A	B	-	-
8.16.2. Test Cable	5						2b	2b	-	-
8.17. Specialized Tools	5						A	A	-	-
9. SWITCHING AND ROUTING TR: 1D7XX Learning Program (Percipio)										
9.1. Switching										
9.1.1. LAN Technologies	5						B	B	-	-
9.2. Virtual Local Area Network (VLAN)										
9.2.1. VLAN Fundamentals	5						B	B	-	-
9.3. Routing										
9.3.1. Routing Fundamentals	5/7						B	B	B	-
10. APPLIANCES (SERVERS) TR: 1D7XX Learning Program (Percipio)										
10.1. Server Types	5						A	A	-	-
10.2. Virtualization	5						B	B	-	-
10.3. Active Directory										
10.3.1. Concepts	5						B	B	-	-

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
10.4. Domain Name System (DNS)										
10.4.1. Concepts	5						B	B	-	-
10.5. Dynamic Host Configuration Protocol (DHCP)										
10.5.1. Concepts	5						B	B	-	-
11. SERVICES AND PROCESSES TR: 1D7XX Learning Program (Percipio)										
11.1. Services										
11.1.1. Services Concepts	5						B	B	-	-
11.1.2. Start / Stop Services							1b	-	-	-
11.2. Processes										
11.2.1. Processes Concepts	5						B	B	-	-
11.2.2. Start / Stop Processes							1b	-	-	-
12. CLIENT SYSTEMS/TROUBLESHOOTING TR: 1D7XX Learning Program (Percipio)										
12.1. End User Devices / Components	5/7						B	B	B	-
12.2. Client Software										
12.2.1. Operating System (OS)	5						A	B	-	-
12.2.2. Mobile Device Operating System	5						A	A	-	-
12.2.3. Software Management Policies	5						A	A	-	-
12.2.4. Install	5						2b	A	-	-
12.3. Scripting										
12.3.1. Conditional Primitives	5						B	B	-	-
12.3.2. Iterative Primitives	5						B	B	-	-
12.3.3. Data Manipulation	5						B	B	-	-
12.3.4. Perform Basic Scripting	5						2b	b	-	-
12.4. Transport Security										
12.4.1. Transport Layer Security	5						-	B	-	-
12.4.2. Encoding	5						-	B	-	-
12.4.3. Tunneling	5						A	B	-	-
12.5. Input Validation										
12.5.1. Data Validation	5						2b	b		
12.5.2. Sanitization	5						2b	b		
12.5.3. SQL Injection	5						-	B		
12.5.4. Code Injection	5						-	B		
12.5.5. Cross Site Scripting	5						-	B		
12.6. Common Protocols (e.g. Handshake, State, OSI Layer, Header Standard Port #)										
12.6.1. SSH	5						A	B	-	-
12.6.2. SSL / TLS	5						A	B	-	-

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12.6.3. Secure FTP Versions	5						A	B	-	-
12.6.4. HTTP(S)	5						A	B	-	-
12.6.5. SNMP	5						A	B	-	-
12.7. Command Line Network Utilities										
12.7.1. Ping	5						A	B	-	-
12.7.2. Tracer / Traceroute	5						A	B	-	-
12.7.3. Ipconfig / Ifconfig	5						A	B	-	-
12.7.4. Netstat	5						A	B	-	-
12.7.5. Nslookup	5						A	B	-	-
13. DATA ANALYST CORE COMPETENCIES										
TR: 1D7X1K Learning Program (Percipio), AFMAN 33-396										
13.1. Operationalized Knowledge Management										
13.1.1. Purpose	5						A	B	-	-
13.2. Agile Learning										
13.2.1. Best Practice Identification / Sharing	5						A	B	-	-
13.2.2. Change Management	5						A	B	-	-
13.2.3. Knowledge Capture	5/7						A	B	B	-
13.2.4. Lessons Learned	5						A	B	-	-
14. WORK CENTER MANAGEMENT										
TR: 1D7XX Learning Program (Percipio); AFIs 10-201, 21-103, 146352; DODI 1400.25V610_AFI36-807; TO 00-33A-1001; AFJQS XXXXX-212S										
14.1. Management Policies	5/7						-	A	A	-
14.2. Training										
14.2.1. Base / Unit Roles and Responsibilities	5/7						-	A	B	-
14.2.2. Supervisor / Trainer Roles and Responsibilities	5/7						-	A	B	-
14.2.3. Task Certifier Roles and Responsibilities	5/7						-	A	B	-
14.2.4. Trainee Responsibilities	5						-	A	-	-
14.2.5. Training Resources										
14.2.5.1. Common Cyber Training Sources	5/7						-	A	B	-
14.2.5.2. 1D7XX/5CXXX and AFS-Specific Learning Programs	5/7						-	A	B	-
14.3. Inspection and Evaluation Programs										
14.3.1. Space Force and Cyber Inspections	5/7/E						-	A	B	B
14.3.2. Self Assessment Program	5/7/E						-	A	B	B
14.4. Automated Information Systems (AIS)										
14.4.1. Job Data / Configuration Management Documentation (e.g. IMDS, LIONS, Remedy, CIPS, IAO Express)	5/7						-	A	B	-

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14.4.2. Training Record Management (e.g. TBA, AFTR, Patriot Excalibur, IKE)	7						-	-	B	-
15. FUNCTIONAL MANAGEMENT TR: 1D7XX Learning Program (Percipio); AFECD; AFI 36-2651; AFPDs 36-28, 38-101; AFMAN 33-396; AFQTP 1D7XX-225E; 1D7XX CFETP										
15.1. Career Field Functional Management	5/7/E						-	A	B	C
15.2. Field Command Functional Management	5/7/E						-	A	B	C
16. RESOURCE MANAGEMENT TR: 1D7XX Learning Program (Percipio); AFPDs 10-6, 65-6; AFIs 10-601, 65601 V(1) and V(2)										
16.1. Financial Management	5/7/E						-	A	B	C
16.2. Funded Requirements	5/7/E						-	A	B	C
16.3. Unfunded Requirements	5/7/E						-	A	B	C
16.4. Resolve Capstone Scenario Involving Resource Management	E						-	-	-	3c
17. MANPOWER AND ORGANIZATION TR: 1D7XX Learning Program (Percipio), AFPD 38-1, AFIs 38-101										
17.1. Manpower Requirements	7/E						-	-	B	C
17.2. Air Force Manpower Standard (AFMS) Application	7/E						-	-	A	B
17.3. Manpower Studies	7/E						-	-	A	B
17.4. Manpower Products	7/E						-	-	B	D
17.5. Allocating Personnel	7/E						-	-	B	D
17.6. Total Force Management	7/E						-	-	B	C

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
1. Organizational Foundation										
1.1. USSF Organization/Administration										
1.1.1. ADCON							A	-	-	-
1.1.2. OPCON							A	-	-	-
1.1.3. TACON							A	-	-	-
1.2. USSF Cyber Units										
1.2.1. Delta 6							A	-	-	-
1.2.2. CDCC-S							A	-	-	-
1.2.3. Cyber Squadrons							A	-	-	-
1.3 USSF DCO Environment										
1.3.1 Defensive Cyber Operations										
1.3.1.1 Organization	5						-	B	-	-
1.3.1.2 Crew Structure	5						-	B	-	-
1.3.1.3 Purpose	5						-	B	-	-
1.3.2 External Elements (Non-USSF)										
1.3.2.1 616 Operations Center							A	-	-	-
1.3.2.2 33 NWS (ACD)							A	-	-	-
1.3.2.3 Cyber Protection Teams (CPT)	5						-	B	-	-
2. Governing Authoritative and Administrative Guidance Block 6 - Introduction to Cyber Warfare Operations, CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918, 1B CFETP (Ch. 5)										
2.1 Law and Ethics										
2.1.1 US Codes (Titles 10, 15, 17, 18, 32, 50)	3						B	-	-	-
2.2 Rules of Engagement (ROE)										
2.2.1 Policy	3						A	-	-	-
2.2.2 Security Tools	3						B	-	-	-
2.2.3 Cyber Management Ethics	3						A	-	-	-
2.2.4 System Monitoring	3						B	-	-	-
2.2.5 Special Data Protection (i.e., Sensitive Personnel Information)	3						A	-	-	-
2.3 Policy and Law										
2.3.1 Executive Orders	3						A	-	-	-
2.3.2 International Laws Affection Electronic Communications	3						A	-	-	-
2.4 U.S. Law										
2.4.1 Military Law	3						A	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
2.4.2 Intellectual Property Laws	3						A	-	-	-
2.4.3 U.S. Law Specific to Electronic Crimes	3						A	-	-	-
2.5 Military Policies										
2.5.1 Operational Risk Management							A	-	-	-
2.5.2 JP 3-12, Cyberspace Operations							B	-	-	-
2.5.3 AFI 17-201 v1-3	5/7/E						A	-	-	-
2.5.4 AFI 17-2 v1-3	5/7/E						A	-	-	-
2.5.5 DCO-S SCG							A	-	-	-
2.5.6 CDCC-S Incident Response Plan							A	-	-	-
2.5.7 DCO CONOPS							B	-	-	-
2.5.8 DCO Pre-Approved Actions							B	-	-	-
2.5.9 Special Instructions (SPINS)							A	-	-	-
2.5.10 Cyber Space Tasking Order (CSTO)							B	-	-	-
2.5.11 Chairman of the Joint Chief of Staff Manual (CJCSM) 6510.01b							A	-	-	-
2.5.12 Ready Cybercrew Program (RCP) Tasking Memorandum (RTM)							B	-	-	-
3. Cyber Operations D Shred Concepts and Fundamentals (CWO Block 6, Introduction to Cyber Warfare Operations), CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT,										
3.1 Fundamentals										
3.1.1 Doctrine, Policy, TTPs and Guidance	3						B	-	-	-
3.1.2 National Strategy	3						A	-	-	-
3.1.3 Command and Control	3						A	-	-	-
3.1.4 Department of Defense Information Network (DoDIN)	3						B	-	-	-
3.1.5 Cyber Organizations and Missions	3						B	-	-	-
3.2 Cyber Mission Force										
3.2.1 Definitions, Roles, and Responsibilities	3						A	-	-	-
3.2.2 Command and Control	3						A	-	-	-
3.3 Mission Areas										
3.3.1 Defend the Nation	3						A	-	-	-
3.3.2 Operate and Defend the DoDIN	3						B	-	-	-
3.3.3 Combatant Command Support	3						B	-	-	-
3.4 Mission Forces										

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL /	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
3.4.1 National Mission Force	3						B	-	-	-
3.4.2 Protection Force	3						B	-	-	-
3.4.3 Combat Mission Force	3						B	-	-	-
3.4.4 Cyber Warfare Operations	3						B	-	-	-
3.4.5 Information Operations	3						B	-	-	-
3.5 Threat Types										
3.5.1 Internal	3						B	-	-	-
3.5.2 External	3						B	-	-	-
3.5.3 State Sponsored	3						B	-	-	-
3.5.4 Non-State Sponsored	3						B	-	-	-
3.6 Cyber Operations										
3.6.1 DoD Information Global Network Operations (DoDIN Ops)	3						B	-	-	-
3.6.2 Defensive Cyber Operations (DCO)										
3.6.2.1 Defensive Cyber Operations-Response Actions (DCO-RA)	3						B	-	-	-
3.6.2.2 Defensive Cyber Operations-Internal Defense Measures (DCO-IDM)	3						B	-	-	-
3.6.2.3 Offensive Cyberspace Operations (OCO)	3						B	-	-	-
3.6.2.4 Legality and Authority	5						B	-	-	-
3.6.2.5 Military and intelligence (Title 10 and Title 50)	5						B	-	-	-
3.7 OCO/DCO Theory and Methodology										
3.7.1 Offensive Theory	3						A	-	-	-
3.7.2 Defensive Theory	3						A	-	-	-
3.7.3 Offensive Methodology	3						A	-	-	-
3.7.4 Defensive Methodology	3						A	-	-	-
3.8 Offensive Methods										
3.8.1 Buffer Overflow Tactics and Techniques	3						2b	-	-	-

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
3.8.2 Privilege Escalation	3						2b	-	-	-
3.8.3 Rootkits	3						2b	-	-	-
3.8.4 Redirection and Triggering	3						2b	-	-	-
3.8.5 Tunneling	3						3b	-	-	-
3.8.6 Collection/Exfiltration	3						2b	-	-	-
3.8.7 Social/Engineering	3						B	-	-	-
3.8.8 Web Exploitation	3						2b	-	-	-
3.8.9 Persistent Access	3						2b	-	-	-
3.8.10 Man-in-the-Middle	3						2b	-	-	-
3.8.11 (Distributed) Denial of Service	3						2b	-	-	-
3.8.12 Obfuscation	3						2b	-	-	-
3.9 Defensive Methods										
3.9.1 Encryption	3						2b	-	-	-
3.9.2 Secure Configurations	3						2b	-	-	-
3.9.3 Secure Enclaves	3						2b	-	-	-
3.9.4 Vulnerability Scans	3						2b	-	-	-
3.9.5 Boundary Protection	3						2b	-	-	-
3.9.6 Intrusion Detection/Prevention (Host/Network)	3						2b	-	-	-
3.10 Active Defense										
3.10.1 Networks										
3.10.1.1 Data	3						B	-	-	-
3.10.1.2 Voice (VoIP, PSTN)	3						B	-	-	-
3.10.1.3 Wireless	3						B	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
3.10.1.4 Space Networks	3						B	-	-	-
3.10.1.5 Industrial Control Systems (ICS)/Supervisory Control and Data Acquisition (SCADA)	3						B	-	-	-
3.10.2 Host Systems										
3.10.2.1 Workstations	3						B	-	-	-
3.10.2.2 Servers	3						B	-	-	-
3.10.3 Web Based										
3.10.3.1 Websites	3						B	-	-	-
3.10.3.2 Databases	3						B	-	-	-
3.10.3.3 Recognize Indicators of Compromise (IOC)	3						2b	-	-	-
3.10.4 Incident Response										
3.10.4.1 Concepts and Methodology							B	-	-	-
3.10.4.2 Incident Response	3						2b	-	-	-
3.10.4.3 Forensics	3						2b	-	-	-
3.10.4.4 Investigation	3						A	-	-	-
3.10.4.5 Evidence Gathering	3						2b	-	-	-
3.10.4.6 Incident Categories	3						A	-	-	-
3.10.4.7 Reporting	3						B	-	-	-
3.11 Information Assurance										
3.11.1 Roles and Responsibilities	3						A	-	-	-
3.11.2 Emissions Security	3						A	-	-	-
3.11.3 Communications Security	3						A	-	-	-
3.11.4 Computer Security	3						A	-	-	-
3.11.5 Operations Security	3						A	-	-	-
3.11.6 Physical Security	3						A	-	-	-

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1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
3.15.1.1 Data	3						B	-	-	-
3.15.1.2 Voice (VoIP, PSTN)	3						B	-	-	-
3.15.1.3 Wireless	3						B	-	-	-
3.15.1.4 Space Networks	3						B	-	-	-
3.15.1.5 Industrial Control Systems (ICS)/Supervisory Control and Data Acquisition (SCADA)	3						B	-	-	-
3.15.2 Host Systems										
3.15.2.1 Workstations	3						B	-	-	-
3.15.2.2 Servers							B	-	-	-
3.15.3 Web Based										
3.15.3.1 Websites	3						B	-	-	-
3.15.3.2 Databases	3						B	-	-	-
4. Conduct OCO/DCO Block 8, Offensive Cyber Operations, Block 9 (Defensive Cyber Operations) Block 11 (Capstone) Block 10 (Ops Planning), CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918, 1B CFETP (Ch 8										
4.1 Planning and Briefing										
4.1.1 Execution and Planning										
4.1.1.2 Pre-Operation Research and Preparation	3						2b	-	-	-
4.1.1.3 Conduct Pre-Mission Actions	3						2b	-	-	-
4.1.1.4 Conduct Post-Mission Actions	3						2b	-	-	-
4.1.1.5 Provide Input to Post Mission Planning	3						2b	-	-	-
4.1.1.6 Measures of Performance/Measures of Effectiveness (MoP/MoE)	5						-	B	-	-
4.1.1.7 Constraints and Restraints	5						-	B	-	-
4.1.2 Prepare Mission Briefs/Debriefs										
4.1.2.1 Provide Input to Pre-Mission Planning	3						2b	-	-	-
4.1.2.2 Provide Input to Mission Debriefs	3						2b	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.1.2.3 Prepare Action Reports	5						-	2b	-	-
4.1.2.4 Assessing Intelligence and Technical Gain/Loss	7						-	-	2c	-
4.1.2.5 Conducting Targeting	7						-	-	2b	-
4.2 Intelligence										
4.2.1 Conduct Open-Source Data Collection	5						-	2b	-	-
4.2.2 Process Exfiltration Data for Analysis	5						-	2b	-	-
4.3 OCO										
4.3.1 Reconnaissance										
4.3.1.1 Network Enumeration										
4.3.1.1.1 Run an Enumeration	3						2b	-	-	-
4.3.1.1.2 Minimize/Obfuscate Network Mapping and Foot Printing	3						B	-	-	-
4.3.1.1.3 Utilize Established SSH Tunnel (Proxychains)	5						-	2b	-	-
4.3.1.2 Host Enumeration										
4.3.1.2.1 Vulnerability Assessment							B	-	-	-
4.3.1.2.2 OS (Version, Patch, etc.)	3						2b	-	-	-
4.3.1.2.3 Ports (Applications and Running Services)	3						2b	-	-	-
4.3.1.2.4 Identify OS's Unique TTL for Packets	3						2b	-	-	-
4.3.2 Exploitation										
4.3.2.1 Determine Payload	3						2b	-	-	-
4.3.2.2 Configure Payload	3						2b	-	-	-
4.3.2.3 Gain Access	3						2b	-	-	-
4.3.3 Post-Exploitation										
4.3.3.1 Establish Persistence							2b	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.3.3.2 Privilege Escalation	3						2b	-	-	-
4.3.3.3 Obtain System Information (Collection of Running Processes, Connected Hosts, Services, etc.)	3						2b	-	-	-
4.3.3.4 Obtain Accounts and Passwords	3						2b	-	-	-
4.3.3.5 Defense Evasion	5						-	2b	-	-
4.3.3.6 Pivoting										
4.3.3.6.1 Lateral Movement	3						2b	-	-	-
4.3.3.6.2 Tunneling	3						2b	-	-	-
4.3.3.6.3 Exfiltration	3						2b	-	-	-
4.3.3.6.4 Password Cracking	3						2b	-	-	-
4.3.4 Password Cracking	3						2b	-	-	-
4.3.5 Wireless										
4.3.5.1 Identify Nearby Wireless Networks	3						2b	-	-	-
4.3.5.2 Crack Session	3						2b	-	-	-
4.3.5.3 Establish Connection to Network	3						2b	-	-	-
4.3.6 TTPs										
4.3.6.1 Create TTPs										
4.3.6.1.1 Enumeration (Discovery)	5						-	2b	-	-
4.3.6.1.2 Exploitation (Execution)	5						-	2b	-	-
4.3.6.1.3 System Access (Initial Access)	5						-	2b	-	-
4.3.6.1.4 Privilege Escalation	5						-	2b	-	-
4.3.6.1.5 Establish Persistence	5						-	2b	-	-
4.3.6.1.6 Credential Access	5						-	2b	-	-
4.3.6.1.7 Defense Evasion	5						-	2b	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.3.6.1.8 Gathering System Information/Files to Exfiltrate (Collection and Staging)	5						-	2b	-	-
4.3.6.1.9 Lateral Movement	5						-	2b	-	-
4.3.6.1.10 Exfiltration	5						-	2b	-	-
4.3.6.1.11 System Impact	5						-	B	-	-
4.3.6.1.12 Evaluate TTPs	7						-	-	D	-
4.3.6.1.13 Provide guidance in absence of established guidance for real-time operations	7						-	-	2b	-
4.3.7 Generate Effects										
4.3.7.1 Generate Denial or Manipulation Effects										
4.3.7.1.1 Networks										
4.3.7.1.1.1 Data							A	-	-	-
4.3.7.1.1.2 Voice (VoIP, PSTN, etc)							A	-	-	-
4.3.7.1.1.3 Wireless							A	-	-	-
4.3.7.1.1.4 Space Networks							A	-	-	-
4.3.7.1.1.5 Industrial Control Systems (ICS)/Supervisory Control and Data Acquisition (SCADA)							A	-	-	-
4.3.7.1.2 Host Systems										
4.3.7.1.2.1 Workstations							A	-	-	-
4.3.7.1.2.2 Servers							A	-	-	-
4.3.7.1.3 Web Based										
4.3.1.3.1 Websites							A	-	-	-
4.3.1.3.2 Databases							A	-	-	-
4.4 DCO										
4.4.1 Weapon System Knowledge										
4.4.1.1 Weapon System (Ex: CVA/H, Manticore, etc.)	5						-	B	-	-
4.4.1.2 Applications	5						-	B	-	-

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.4.1.3 Kit Management System [Ex: tPlenum (VM)]										
4.4.1.4 Security Event and Incident Management (SEIM)										
4.4.1.4.1 Dashboard Management/Navigation (Ex: Kibana)										
4.4.1.4.1.1 Edit/Customize Dashboard	5						-	2b	-	-
4.4.1.4.1.2 Create a New Alert	5						-	2b	-	-
4.4.1.4.2 Create Incident Case (Ex: TheHive)	5						-	2b	-	-
4.4.1.4.3 Create Network Rule (Ex: Snort, Suricata)	5						-	2b	-	-
4.4.2 Scheduling Platforms (Ex: Project IKE, PEX, etc.)										
4.4.2.1 Dashboard Management/Navigation										
4.4.2.1.1 Edit/Customize Dashboard	3						2b	-	-	-
4.4.2.1.2 Change View/Spaces	3						2b	-	-	-
4.4.2.2 Update Personnel Records										
4.4.2.2.1 Add/Remove Members from unit	5						-	2b	-	-
4.4.2.2.2 Update Training Records	5						-	2b	-	-
4.4.2.3 Updating Terrain	5						-	2b	-	-
4.4.3 Active Monitoring										
4.4.3.1 Alert Triage			3						B	-
4.4.3.2 Log/Alert Enrichment			3						B	-
4.4.3.3 Case Management	3						B	-	-	-
4.4.3.4 Signature Management	3						B	-	-	-
4.4.3.5 Rule Logic Types	3						B	-	-	-
4.4.3.6 Malicious Use of Common Protocols	3						B	-	-	-
4.4.3.7 Analysis Bias	3						B	-	-	-
4.4.3.8 Baselineing	3						B	-	-	-

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.4.4 Incident Response										
4.4.4.1 Methodology	5						-	B	-	-
4.4.4.2 Event/Incident Categorization	5						-	B	-	-
4.4.4.3 Remote Evidence Collection	5						-	B	-	-
4.4.4.4 Reporting	5						-	B	-	-
4.4.4.5 Forensics	5						-	B	-	-
4.4.4.5.1 Log Files	3						B	-	-	-
4.4.4.6 Conduct Investigation	5						-	B	-	-
4.4.4.7 Incident Recovery	5						-	B	-	-
4.4.4.8 Containment Techniques	3						B	-	-	-
4.4.4.9 Eradication Techniques							B	-	-	-
4.4.4.9.1 Block Domains	3						2b	-	-	-
4.4.4.9.2 Block IP Addresses	3						2b	-	-	-
4.4.4.9.3 Block Accounts	3						2b	-	-	-
4.4.4.9.4 File Removal	3						2b	-	-	-
4.4.4.9.5 Kill Processes	3						2b	-	-	-
4.4.4.10 Common Attack Methods vs Mitigations	3						B	-	-	-
4.4.4.11 Analysis Techniques	5						-	2c	-	-
4.4.4.12 Assess and Recommend Remediation Actions	5						-	2c	-	-
4.4.5 Forensics										
4.4.5.1 Network							-	-	-	-
4.4.5.1.1 Source Data Types							-	-	-	-
4.4.5.1.1.1 Full-Packet Capture	3						2b	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.4.5.1.1.2 NetFlow & Related Flow-Based Collections	3						B	-	-	-
4.4.5.1.1.3 Log Files	3						B	-	-	-
4.4.5.1.2 Network Traffic Anomalies	3						B	-	-	-
4.4.5.1.3 Network Log Analysis (Ex: Zeek)	5						B	-	-	-
4.4.5.2 Host							-	-	-	-
4.4.5.2.1 Data Acquisition							-	-	-	-
4.4.5.2.2 Event Log Analysis	5						-	2c	-	-
4.4.5.2.3 Host Memory Analysis	5						-	2c	-	-
4.4.5.2.4 File System Analysis	5						-	2c	-	-
4.4.5.2.5 Timeline Analysis	5						-	2c	-	-
4.4.5.2.6 Disk Forensics							-	-	-	-
4.4.5.2.7 Logs							-	-	-	-
4.4.5.2.7.1 Windows							-	-	-	-
4.4.5.2.7.1.1 Windows Event Viewer	3						B	-	-	-
4.4.5.2.7.1.2 Windows Log Location (C:\Windows\system32\config) (							-	-	-	-
4.4.5.2.7.2 *NIX							-	-	-	-
4.4.5.2.7.2.1 Log Types							-	-	-	-
4.4.5.2.7.2.1.1 Binary	3						B	-	-	-
4.4.5.2.7.2.1.2 System	3						B	-	-	-
4.4.5.2.7.2.2 Common Location (/var/log)	3						B	-	-	-
4.4.5.2.7.2.3 Key Logs	3						B	-	-	-
4.4.5.2.7.2.3.1 Last Login (/var/log/lastlog)							-	-	-	-
4.4.5.2.7.2.3.2 Failed Login (/var/log/btmp)							-	-	-	-
4.4.5.2.7.2.3.3 Historic Logins (/var/log/wtmp)							-	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
4.4.5.2.7.2.3.4 Sudo History (/etc/sudoers)							-	-	-	-
4.4.5.2.8 Autoruns							-	-	-	-
4.4.5.2.8.1 Windows							-	-	-	-
4.4.5.2.8.1.1 Registry	3						B	-	-	-
4.4.5.2.8.1.2 Scheduled Tasks	3						B	-	-	-
4.4.5.2.8.2 *NIX							-	-	-	-
4.4.5.2.8.2.1 Crontab	3						B	-	-	-
4.4.5.2.9 Hidden Files							-	-	-	-
4.4.5.2.9.1 Windows Alternate Data Stream (ADS)	3						B	-	-	-
4.4.6 Reporting							-	-	-	-
4.4.6.1 Cyber 9-Line	5						-	B	-	-
4.4.6.2 Reporting Times	5						-	B	-	-
4.4.6.3 MISREP	5						-	B	-	-
4.4.6.4 Impact Assessment Matrix	5						-	B	-	-
5 Networking Concepts Block 4 (Network Fundamentals) Block 5 (Network Configuration) 1B CFETP (Ch 2), CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918										
5.1 Digital Numbering Systems (Internal Data Representation)										
5.1.1 Binary	3						A	-	-	-
5.1.2 Hexadecimal	3						A	-	-	-
5.2 Standards and Framework										
5.2.1 Structure							-	-	-	-
5.2.1.1 OSI Model	3						B	-	-	-
5.2.1.2 TCP/IP Suite	3						B	-	-	-
5.2.2 Protocol Data Unit							-	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
5.2.2.1 Transmission Control Protocol (TCP)	3						B	-	-	-
5.2.2.2 User Datagram Protocol (UDP)	3						B	-	-	-
5.2.3 Encapsulation/Decapsulation	3						B	-	-	-
5.3 Networking Addressing										
5.3.1 Data-Link Layer							-	-	-	-
5.3.1.1 Media Access Control (MAC)							-	-	-	-
5.3.1.1.1 Address Structure	3						B	-	-	-
5.3.1.1.2 Ethernet Frame Structure	3						B	-	-	-
5.3.2 Network Layer							-	-	-	-
5.3.2.1 Internet Protocol (v4 & v6)							-	-	-	-
5.3.2.1.1 Address Structure	3						B	-	-	-
5.3.2.1.2 Packet Structure	3						B	-	-	-
5.3.2.1.3 Address Types							-	-	-	-
5.3.2.1.3.1 Classful	3						B	-	-	-
5.3.2.1.3.2 Classless	3						B	-	-	-
5.3.2.1.3.3 Private/Public	3						B	-	-	-
5.3.2.1.3.4 Reserved	3						B	-	-	-
5.3.2.1.4 IPv4 Subnetting	3						B	-	-	-
5.4 Routing Protocols										
5.4.1 Interior	3						A	-	-	-
5.4.2 Exterior	3						A	-	-	-
5.4.3 Link-State	3						A	-	-	-
5.5 Networks										
5.5.1 Topologies (Star, Ring, Bus, Hybrid, etc.)	3						A	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
5.5.2 Virtual Private Network (VPN)	3						A	-	-	-
5.5.2.1 Protocols	3						A	-	-	-
5.5.2.2 Components	3						A	-	-	-
5.5.3 Segmentation & Compartmentalization							A	-	-	-
5.5.3.1 Virtual Local Area Network (VLAN)	3						A	-	-	-
5.5.3.1.1 Concepts	3						A	-	-	-
5.5.3.1.2 Private VLANs	3						A	-	-	-
5.5.4 Wireless							-	-	-	-
5.5.4.1 Topology	3						A	-	-	-
5.5.4.2 Components	3						A	-	-	-
5.5.4.3 Security	3						A	-	-	-
5.5.5 Network Address Translation/Port Address Translation (NAT/PAT)							A	-	-	-
5.5.5.1 Concepts/Fundamentals	3						A	-	-	-
5.6 Protocols										
5.6.1 Concepts	3						A	-	-	-
5.6.2 Commonly Used Protocols (DHCP, STMP, DNS)	3						A	-	-	-
5.7 Ports										
5.7.1 Concepts	3						A	-	-	-
5.7.2 Commonly Used Ports	3						A	-	-	-
5.8 Network Traffic Analysis										
5.8.1 Fundamentals	3						A	-	-	-
5.8.2 Analysis Process							-	-	-	-
5.8.2.1 Capture Traffic	3						2b	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
5.8.2.2 Analyze Traffic	3						2c	-	-	-
5.8.2.3 Identify Encoded Traffic	3						2b	-	-	-
5.8.2.4 Identify Malicious Traffic	3						2b	-	-	-
5.8.2.5 Identify the OS version from Network Traffic	3						2b	-	-	-
5.8.2.6 Identify the patch release of OS software from Network Traffic	3						2b	-	-	-
5.8.2.7 Identify Services and Applications on a Network	3						2b	-	-	-
6 Network Configuration Block 5 (Network Configuration), 1B CFETP (Ch 2), CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918										
6.1 Device Configuration										
6.1.1 Cisco										
6.1.1.1 Navigate Cisco IOS CLI	3						2b	-	-	-
6.1.1.2 Configuration							B	-	-	-
6.1.1.2.1 Start-up	3						2b	-	-	-
6.1.1.2.2 Running	3						2b	-	-	-
6.1.1.2.3 Management							B	-	-	-
6.1.1.2.3.1 Write (Copy Run Start)	3						2b	-	-	-
6.1.1.2.3.2 Erase	3						2b	-	-	-
6.1.1.3 Router										
6.1.1.3.1 Port Configuration	3						B	-	-	-
6.1.1.3.2 Configure Routing Protocol							2b	-	-	-
6.1.1.3.2.1 Internal Routing Protocol (IRP)	3						B	-	-	-
6.1.1.3.2.2 External Routing Protocol (ERP)	3						B	-	-	-
6.1.1.3.3 Apply Access Control List							2b	-	-	-
6.1.1.3.3.1 Standard	3						B	-	-	-
6.1.1.3.3.2 Extended	3						B	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
6.1.1.4 Switch										
6.1.1.4.1 Port Configuration	3						B	-	-	-
6.1.1.4.1.1 Switchport							B	-	-	-
6.1.1.4.1.1.1 Access	3						B	-	-	-
6.1.1.4.1.1.2 Trunk	3						B	-	-	-
6.1.1.4.1.1.2.1 Native VLAN	3						B	-	-	-
6.1.1.4.1.1.3 Portfast	3						B	-	-	-
6.1.1.4.1.2 No Switchport	3						B	-	-	-
6.1.1.4.1.2.1 IP Configuration	3						B	-	-	-
6.1.1.4.1.3 Security							B	-	-	-
6.1.1.4.1.3.1 MAC-Address	3						B	-	-	-
6.1.1.4.1.4 SPAN Port	3						B	-	-	-
6.1.1.4.2 Apply Access Control List							2b	-	-	-
6.1.1.4.2.1 Standard	3						B	-	-	-
6.1.1.4.2.2 Extended	3						B	-	-	-
6.1.1.4.3 Switched Virtual Interface (SVI)	3						B	-	-	-
6.1.1.4.4 VLAN							B	-	-	-
6.1.1.4.4.1 Database Configuration	3						B	-	-	-
6.1.1.4.4.2 Interface Configuration	3						B	-	-	-
6.1.1.4.5 IP Routing	3						B	-	-	-
7 Operating Concepts Block 1 (Windows), Block 2 (Linux), 1B CFETP (Ch. 3), CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Work roles (CMT, CST, NMT, NST) 20200918										
7.1 Concepts										
7.1.1 Boot Process							A	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
7.1.1.1 BIOS	3						A	-	-	-
7.1.1.2 OS Specific							A	-	-	-
7.1.1.2.1 Windows Boot Manager	3						A	-	-	-
7.1.1.2.2 GRUB (Stage 1 and 2)	3						A	-	-	-
7.1.1.2.3 Init	3						A	-	-	-
7.1.2 Kernel	3						A	-	-	-
7.1.3 Drivers	3						A	-	-	-
7.2 Format										
7.2.1 System Formats							A	-	-	-
7.2.1.1 FAT	3						A	-	-	-
7.2.1.2 exFAT	3						A	-	-	-
7.2.1.3 NTFS	3						A	-	-	-
7.2.1.3.1 Security Benefits	3						A	-	-	-
7.2.2 Differences	3						A	-	-	-
7.2.3 File Formats (e.g., EXE, PDF, PCAP)	3						A	-	-	-
7.2.4 Compression (e.g., ZIP, TAR, 7z)	3						A	-	-	-
7.3 Windows										
7.3.1 Components							A	-	-	-
7.3.1.1 Registry	3						A	-	-	-
7.3.2 File Structure	3						A	-	-	-
7.3.3 Utilize CLI	3						A	-	-	-
7.3.3 Utilize GUI	3						A	-	-	-
7.3.4 Manipulate System							A	-	-	-
7.3.4.1 User Accounts	3						A	-	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
7.3.4.2 File Systems	3						A	-	-	-
7.3.4.3 Network Shares	3						A	-	-	-
7.3.4.4 Network Settings	3						A	-	-	-
7.3.4.5 Services	3						A	-	-	-
7.3.4.6 Firewall	3						A	-	-	-
7.3.4.7 Logs	3						A	-	-	-
7.3.5 Utilize Registry	3						A	-	-	-
7.4 *NIX										
7.4.1 Components							A	-	-	-
7.4.1.1 /proc	3						A	-	-	-
7.4.1.2 Shell							A	-	-	-
7.4.1.2.1 Types (Shell, Bash, zShell, etc.)	3						A	-	-	-
7.4.1.2.2 Understand Certain Limitations and Differences	3						A	-	-	-
7.4.2 File Structure	3						A	-	-	-
7.4.3 Utilize CLI	3						A	-	-	-
7.4.4 Utilize GUI	3						A	-	-	-
7.4.5 Linux-Based Distros	3						A	-	-	-
7.4.6 Manipulate System							A	-	-	-
7.4.6.1 User Accounts	3						A	-	-	-
7.4.6.2 File Systems	3						A	-	-	-
7.4.6.3 Network Shares	3						A	-	-	-
7.4.6.4 Network Settings	3						A	-	-	-
7.4.6.5 Services	3						A	-	-	-

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
7.4.6.6 Packages	3						A	-	-	-
7.4.6.7 Logs	3						A	-	-	-
7.4.7 Utilize /proc	3						A	-	-	-
7.4.8 Credentials							A	-	-	-
7.4.9 System Hardening							A	-	-	-
7.7 Scripting										
7.7.1 Fundamentals, Ability to Create or Edit:							-	-	-	-
7.7.1.1 Batch	3						2b	-	-	-
7.7.1.2 PowerShell	3						2b	-	-	-
7.7.1.3 Bash	3						2b	-	-	-
7.7.1.4 Python	3						2b	-	-	-
7.8 Virtualization										
7.8.1 Concepts	3						A	-	-	-
7.8.2 Types	3						A	-	-	-
8 Planning Block 10 (Ops Planning), 1B CFETP (Ch 7, 11, 12), 65 CYS MDT MQT Training Task List, CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918										
8.1 Joint										
8.1.1 Joint Command and Planning Process							-	-	-	-
8.1.1.1 Concepts	3						A	-	-	-
8.1.1.2 Structure and Organization	3						A	-	-	-
8.1.1.3 Levels of War	5						-	A	-	-
8.1.1.4 Roles and Responsibilities	5						-	A	-	-
8.1.2 Command and Control (C2)	5						-	B	-	-
8.1.3 Authorities	5						-	B	-	-

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
8.1.4 Orders	5						-	B	-	-
8.1.5 Planning Process Defined (JPP)							A	-	-	-
8.1.5.1 Concepts	3						A	-	-	-
8.1.5.2 Deliberate Planning	7						-	-	B	-
8.1.5.3 Crisis Action Planning	7						-	-	B	-
8.2 Lines of Operations										
8.2.1 DoDIN	3						B	-	-	-
8.2.2 DCO	3						B	-	-	-
8.2.3 OCO	3						B	-	-	-
8.3 Cyberspace Operational Planning										
8.3.1 Cyber C2	5						-	B	-	-
8.3.2 Synchronization	5						-	B	-	-
8.3.4 Weaponizing	5						-	B	-	-
8.3.5 Asset/Target Analysis	5						-	B	-	-
8.3.6 Intelligence Gain/Loss	5						-	B	-	-
8.3.7 Technical Gain/Loss	5						-	B	-	-
8.3.8 Deconfliction	5						-	B	-	-
8.3.9 Conduct OCO and DCO Mission Planning	5						-	2b	-	-
8.3.9.1 Produce strategy to inform commander's decision-making process	7						-	B	-	-
8.4 Cyber Operations Assessments										
8.4.1 Measures of Performance/Measures of Effectiveness (MoP/MoE)	5						-	B	-	-
8.4.1.1 Create Measures	5						-	2b	-	-
8.4.1.2 Evaluate how the operation was successful or ineffective to objectives							-	-	-	-
8.4.2 Conducting Ops Assessment	5						-	2b	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
8.4.3 Battle Damage Assessment	7						-	-	2b	-
8.5 Mission Planning (PBED)										
8.5.1 Concepts	3						A	-	-	-
8.5.2 Plan							A	-	-	-
8.5.2.1 ME3C-(PC)*2	5						-	B	-	-
8.5.2.2 Mission Planning Cell	5						-	B	-	-
8.5.2.3 Build Mission Plan	5						-	B	-	-
8.5.2.4 Mission Types	5						-	B	-	-
8.5.3 Brief							-	B	-	-
8.5.3.1 Mission Approval Brief	5						-	B	-	-
8.5.3.2 Pre-Sortie Brief	5						-	B	-	-
8.5.4 Execute							-	B	-	-
8.5.4.1 Pre-Mission Tasks	5						-	B	-	-
8.5.4.2 Post-Mission Tasks	5						-	B	-	-
8.5.4.3 Crew Changeover	5						-	B	-	-
8.5.4.4 Operators Notes/Accountability	5						-	B	-	-
8.5.5 Debrief							-	B	-	-
8.5.5.1 Debrief Process	5						-	B	-	-
8.5.5.2 Instructional Fix/Lessons Learned Implementation Process	5						-	B	-	-
8.5.5.3 Participate in debrief	5						-	2b	-	-
9 Intelligence CTM 7-0.1 Joint Cyberspace Trng and Cert Standards (JCT_CS) v4.0 ANNEX B Workroles (CMT, CST, NMT, NST) 20200918, 1B CFETP (Ch. 11, 12, 13)										
9.1 Fundamentals										
9.1.1 Data, Information, and Intelligence (Differences and Relationships)	5						-	B	-	-

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
9.1.2 Identify the Intelligence Process	5						-	B	-	-
9.1.3 Identify the Targeting Cycle	5						-	B	-	-
9.1.4 Intelligence Sources/Disciplines (e.g., OSINT)	5						-	B	-	-
9.1.5 Intelligence Reports (e.g., IIR)	5						-	B	-	-
9.1.6 Request for Information (RFI)	5						-	B	-	-
9.1.7 Tactics from Intel Sources	7						-	-	B	-
9.1.8 Sources							B	-	-	-
9.1.8.1 Open-Source							B	-	-	-
9.1.8.2 Closed-Source							B	-	-	-
9.1.9 Intel Integration (Squadron)							B	-	-	-
9.1.10 Cyber Intelligence Models							B	-	-	-
9.1.11 DEL 7 Det 6							B	-	-	-
9.2 Collection Management										
9.2.1 Complete RFI and FFIR Forms	5						-	2b	-	-
9.2.2 Intelligence Limitations/Problems	5						-	B	-	-
9.2.3 Collection Requirements	5						-	B	-	-
9.2.4 Commander's Critical Information Requirement (CCIR)	7						-	-	B	-
9.2.5 Priority Intelligence Requirements (PIR)	7						-	-	B	-
9.2.6 Essential Elements of Information (EEI)	7						-	-	B	-
9.3 Informed Operations										
9.3.1 Applied to cyberspace operations, identify how intelligence informs:							-	-	-	-
9.3.1.1 Mission Analysis activities	7						-	-	B	-
9.3.1.2 Operational Risk Assessment	7						-	-	B	-
9.3.1.3 Drives the tactical planning and/or operations cycle	7						-	-	B	-

Qualitative Requirements

PROFICIENCY CODE KEY		
	SCALE VALUE	DEFINITION: The individual
Task Performance Levels	1	Can do simple parts of the task. Needs to be told or shown how to do most of the task. (EXTREMELY LIMITED)
	2	Can do most parts of the task. Needs help only on hardest parts. (PARTIALLY PROFICIENT)
	3	Can do all parts of the task. Needs only a spot check of completed work. (COMPETENT)
	4	Can do the complete task quickly and accurately. Can tell or show others how to do the task. (HIGHLY PROFICIENT)
*Task Knowledge Levels	a	Can name parts, tools, and simple facts about the task. (NOMENCLATURE)
	b	Can determine step by step procedures for doing the task. (PROCEDURES)
	c	Can identify why and when the task must be done and why each step is needed. (OPERATING PRINCIPLES)
	d	Can predict, isolate, and resolve problems about the task. (ADVANCED THEORY)
**Subject Knowledge	A	Can identify basic facts and terms about the subject. (FACTS)
	B	Can identify relationship of basic facts and state general principles about the subject. (PRINCIPLES)
	C	Can analyze facts and principles and draw conclusions about the subject. (ANALYSIS)
	D	Can evaluate conditions and make proper decisions about the subject. (EVALUATION)
Explanations * A task knowledge scale value may be used alone or with a task performance scale value to define a level of knowledge for a specific task. (Example: b and 1b) ** A subject knowledge scale value is used alone to define a level of knowledge for a subject not directly related to any specific task, or for a subject common to several tasks. This mark is used alone instead of a scale value to show that no proficiency training is provided in the course or Learning Program. (-) This mark is used alone in the Proficiency Codes Course columns to show no proficiency training is provided in the applicable course. Training is satisfied through OJT, CBTs, Learning Programs, or a combination. (X) This mark is used alone in the Proficiency Codes Course columns to show training is required but not given due to limitations in resources or is a future requirement. Training is satisfied through OJT, CBTs, Learning Programs, or a combination. NOTE: All tasks and knowledge items shown with a proficiency code are trained during wartime. (X) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task as a core requirement to the level identified in the Task Knowledge Levels Column. The training to satisfy this requirement is either provided through OJT, CBTs, Learning Programs, or a combination. (*) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task as a wartime/deployment requirement. (^) When this code is used in the Core & Wartime Tasks Column it indicates the CFM has mandated this task require(s) third person certification.		

CFETP versus AFJQS task coding. AFJQSs/AFQTPs annotated in the CFETP with a skill level denotes the AFJQS is mandatory.

1. Implementation. This STS will be used for technical training provided by AETC for the 3-level course.

1. TASKS, KNOWLEDGE AND TECHNICAL REFERENCES	2. CORE & WARTIME TASKS	3. CERTIFICATION FOR OJT					4. PROFICIENCY CODES USED TO INDICATE TRAINING/INFORMATION PROVIDED - COMPETENCY LEVELS			
		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
1. EXPEDITIONARY COMMUNICATIONS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
1.1. Duties of SFSC	5/7						A	B	B	-
2. SAFETY / RISK MANAGEMENT (RM) TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/); AFIs 32-1065, 90-802, 91-202; AFMAN 91-203; AFPAM 90-803; AFPD 91-2; MIL-STD 188-124B										
2.1. Climbing and Working Aloft							-	-	-	-
2.2. Radio Frequency Radiation	5/7						A	B	C	-
2.3. DoD Electromagnetic Environmental Effects (E3) Program	5/7						-	A	B	-
2.4. Lightning Protection and Grounding	5						A	A	-	-
3. BASIC ELECTRONICS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
3.1. Metric Notation	5/7						A	A	A	-
3.2. Decibel Math (Logarithmic Power Calculation)	5						A	B	B	-
3.3. Prefixes	5						A	A	-	-
3.4. Fundamentals of Electricity	5/7						B	B	C	-
3.5. Component and Device Theory	5/7						A	B	B	-
3.6. Wave Generating Circuits	5						A	A	-	-
3.7. Digital Circuits	5						A	A	-	-
4. CABLE FUNDAMENTALS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
4.1. Wire Color Coding Standards	5/7						A	A	B	-
4.2. Fiber Optic Cable	5						B	B	-	-
4.3. Twisted Pair	5						A	A	-	-
4.4. Coaxial Cable	5						B	B	-	-
4.5. Shielding	5						A	B	-	-
4.6. Labeling	5/7						A	B	B	-
4.7. Build Ethernet Cable	5/7						2b	B	B	-
4.8. Build Coax Cable	5/7						2b	B	B	-
4.9. Coax Testing	5						A	B	-	-
4.10. Perform Cable Testing	5/7						2b	B	B	-
5. LONG HAUL COMMUNICATIONS CONCEPTS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
5.1. Global Circuit and Transport Concepts										
5.1.1. Terrestrial Transport Methods										
5.1.1.1. Enterprise Networks	5/7						B	B	C	-
5.1.1.2. DISA Circuits	5/7						A	A	B	-

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		A	B	C	D	E	3 SKILL LEVEL / BASIC	5 SKILL LEVEL / INTERMEDIATE	7 SKILL LEVEL / ADVANCED	EXPERT LEVEL
		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
5.1.1.3. Commercial ISPs	5/7						A	A	B	-
5.1.2. Wireless Long Haul Fundamentals										
5.1.2.1. LOS	5/7						A	A	B	-
5.1.2.2. SATCOM	5/7						A	A	B	-
5.1.2.3. Troposcatter	5/7						A	A	B	-
6. ENCRYPTION / DECRYPTION TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
6.1. Crypto Keys										
6.1.1. Pre Placed Key (PPK)	5/7						A	A	B	-
6.1.2. Firefly Vector Set	5/7						A	A	B	-
6.2. Key Roles										
6.2.1. Traffic Encryption Keys (TEK)	5/7						A	A	B	-
6.2.2. Key Encryption Keys (KEK)	5/7						A	A	B	-
6.2.3. TrKEK	5/7						A	A	B	-
7. RF TRANSMISSION FUNDAMENTALS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/), AF1 36-2101, 1D7XX CFETP, AFECD										
7.1. Radio Theory										
7.1.1. Transmitters	5						A	A	-	-
7.1.2. Receivers	5						A	A	-	-
7.1.3. Transceivers	5						A	A	-	-
7.1.4. RF Transmission Mediums	5/7						A	B	C	-
7.2. Modulation Techniques										
7.2.1. Understand Modulation	5						B	B	C	-
7.2.2. Amplitude Modulation	5						B	B	-	-
7.2.3. Frequency Modulation	5						B	B	-	-
7.2.4. Digital Modulation	5/7						A	B	C	-
7.3. RF Spectrum										
7.3.1. Frequency Bands and Characteristics	5/7						A	B	C	-
7.4. Electromagnetic Wave Propagation Theory										
7.4.1. Radio Wave Propagation Principles	5/7						B	B	C	-
7.4.2. Refraction	5						A	B	-	-
7.4.3. Reflection	5						A	B	-	-
7.4.4. Diffraction	5						A	B	-	-
7.4.5. Skywave Fundamentals	5/7						A	B	C	-
7.5. Signal Loss										
7.5.1. Path Loss	5						A	B	-	-
7.5.2. Atmospheric Attenuation	5						A	B	-	-
7.5.3. Multipathing	5						A	B	-	-

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
7.5.4. Free Space Loss	5/7						A	B	C	-
7.5.5. Anomalous Propagation	5/7						A	B	C	-
7.5.6. Solar Emissions and Effects	5						A	B	-	-
8. TEST EQUIPMENT TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
8.1. Multimeter	5						2b	B	-	-
8.2. Built-in Test Equipment	5						A	A	-	-
8.3. Communication Systems Analyzer	5						2b	A	-	-
8.4. Dummy Load	5						A	A	-	-
8.5. VSWR Tester	5						A	A	-	-
8.6. Compass / Inclinator	5						2b	A	-	-
9. ANTENNA FUNDAMENTALS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
9.1. Antenna RF Propagation Theory										
9.1.1. Fundamentals of Antenna Propagation	5/7						B	B	C	-
9.1.2. Antenna Gain	5/7						A	B	B	-
9.1.3. Polarization	5/7						A	B	C	-
9.1.4. Mutual Interference	5/7						A	B	B	-
9.1.5. Impedance Matching	5/7						A	B	B	-
9.1.6. Deployable Antenna Types	5/7						A	A	B	-
9.1.7. Beamwidth	5/7						A	A	B	-
9.1.8. Calculate Electrical and Physical Length	5						2b	A	-	-
9.1.9. Relationship of Antenna Height and Takeoff Angle	5						B	B	-	-
9.2. Antenna Types and Applications										
9.2.1. Omnidirectional and Directional Antennas	5						A	B	-	-
9.2.2. Common Antenna Types	5						A	B	-	-
9.2.3. Airborne Antenna Applications	5/7						-	A	B	-
9.2.4. Basic Antenna Tests	5						A	A	-	-
10. SATCOM FUNDAMENTALS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
10.1. Satellite System Segments Principles, Capabilities, and Limitations										
10.1.1. Space Segment	5						A	A	-	-
10.1.2. Command and Control Segment	5						A	A	-	-
10.1.3. Terminal (Ground) Segment	5						A	A	-	-
10.2. Satellite Bands, Purpose, Capabilities, and Limitations										

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
10.2.1. IEEE and ITU Radio Frequency Band Standards	5/7						A	A	B	-
10.2.2. L, C, X, Ku, K, and Ka Band applications and considerations	5/7						A	B	C	-
10.3. Satellite Access Principles										
10.3.1. FDMA	5/7						B	B	B	-
10.3.2. TMDA	5/7						B	B	B	-
10.3.3. CDMA	5/7						A	B	B	-
10.4. Current Satellite Constellations										
10.4.1. SATCOM Terminal Characteristics										
10.4.1.1. Introduction to UHF, SHF, EHF Terminals	5						A	B	-	-
10.4.1.2. Multiband Satellite Terminals	5/7						A	B	B	-
10.4.2. Phase Modulation										
10.4.2.1. Understand Phase Modulation	5/7						B	B	C	-
10.4.2.2. Types of Modulation	5/7						B	B	B	-
10.4.2.3. Modulation Considerations	5/7						A	B	C	-
10.4.2.4. Forward Error Correction (FEC)	5/7						A	B	B	-
10.4.3. Transmit Systems										
10.4.3.1. Modems	5						A	B	-	-
10.4.3.2. Transmitters	5						A	B	-	-
10.4.3.3. Upconverters										
10.4.3.3.1. Upconverter	5						A	B	-	-
10.4.3.3.2. Block Upconverter	5						A	B	-	-
10.4.3.4. Power Amplifier	5						A	B	-	-
10.4.4. Receive Systems										
10.4.4.1. Receiver	5						A	B	-	-
10.4.4.2. Low Noise Amplifier	5						A	B	-	-
10.4.4.3. Low Noise Block	5						A	B	-	-
10.4.4.4. Downconverters										
10.4.4.4.1. Downconverter	5						A	B	-	-
10.4.4.4.2. Block Downconverter TR: subtask of 11.4.4.4	5						A	B	-	-
10.4.4.5. Demodulator	5						A	B	-	-
10.4.5. Antenna Systems										
10.4.5.1. Antenna Systems	5						A	B	-	-
10.4.5.2. Waveguides	5						A	B	-	-
10.4.5.3. Feed Horns / Feed Assemblies	5						A	B	-	-

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		START DATE	STOP DATE	TRAINEE INITIALS	TRAINER INITIALS	CERTIFIER INITIALS	COURSE	COURSE	COURSE	COURSE
10.4.5.4. Acquisition and Tracking Principles	5						A	B	-	-
10.4.5.5. Calculate Satellite Look Angles	5						2b	B	-	-
10.4.5.6. Acquire and Track Satellites	5						2b	B	-	-
10.4.6. Global Positioning System Receivers										
10.4.6.1. Principles, Capabilities, and Limitations	5						A	B	C	-
10.5. SATCOM Link Operations										
10.5.1. Communication Link Establishment and Maintenance										
10.5.1.1. Principles	5/7						A	B	B	-
10.5.1.2. Establish a Communications Link	5/7						2b	B	B	-
10.5.1.3. Counter Counter Measure Principles	5/7						A	B	C	-
10.5.1.4. Maintain Master Station Logs	5/7						A	A	B	-
10.5.1.5. Develop After Action Reports	5/7						A	A	B	-
11. ELECTRICAL POWER SYSTEMS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
11.1. Uninterruptable Power Supplies	5/7						A	A	B	-
11.2. Batteries	5/7						A	A	B	-
11.3. Inverters	5/7						A	A	B	-
11.4. Filters	5/7						A	A	B	-
11.5. Generators	5/7						A	A	B	-
11.6. Considerations for Field Application	5/7						A	B	C	-
12. NETWORK AUTHENTICATION FUNDAMENTALS TR: 1D7X1W Expeditionary Communications (Percipio - https://usaf.percipio.com/)										
12.1. 802.1X	5/7						-	B	B	-
12.2. Port Security	5						-	B	-	-
12.3. PKI	5/7						-	B	B	-
13. ELECTROMAGNETIC SPECTRUM OPERATIONS										
13.1. Localized Interference Resolution	5/7						A	B	C	-
13.2. ELECTROMAGNETIC WARFARE (EW)										
13.2.1. Electromagnetic Attack (EA)	5/7						A	B	C	
13.2.2. Electromagnetic Protection (EP)	5/7						A	B	C	
13.2.3. Electromagnetic Support (ES)	5/7						A	B	C	
13.3. Spectrum Sensing	5/7						-	A	B	-
13.4. Unit / AOR Comm Plans (ANNEX K) PACE	5/7						A	B	C	-